Year 10 Mathematics

Algebra 1

2.3 Simplifying Expressions: Multiplication

Simplifying Expressions: Multiplication

STARTER · 5 MIN

Date: _____ **Title:** Simplifying Expressions: Multiplication

LEARNING OBJECTIVES

- Multiply terms together
- Expand a single bracket
- Expand and simplify expressions with brackets

STARTER *Answer in your book. No calculators.*

1. Work out $4 \times (-3)$
2. Work out $-6 \times (-2)$
3. Work out 5^2
4. Simplify $3x + 4 + 2x - 1$

Starter — Answers

STARTER

Tick or correct in green pen

1. -12
2. 12
3. 25
4. $5x + 3$

Success Criteria

By the end of this lesson:

BRONZE	I can multiply terms together	☐
SILVER	I can expand a single bracket	☐
GOLD	I can expand and simplify expressions with brackets	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Multiplying and Expanding

INTRODUCTION · 3 MIN

Multiply the numbers first, then multiply the letters.

Multiply

Multiply the numbers, then the letters.

$3a \times 4b = 12ab$

Same letter

Use a power for a repeated letter.

$2a \times 3a = 6a^2$

Expand

Multiply every term inside the bracket.

$3(x + 4) = 3x + 12$

Simplify

Expand, then collect like terms.

$2(x + 1) + x = 3x + 2$

Examples 1 and 2 — Multiplying Terms

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Simplify

$3x \times 5y$

Example 2

Simplify

$2a \times 4a$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$3 \times 5 = 15$

$x \times y = xy$

$= 15xy$

Tip: write the letters in alphabetical order.

Example 2

$2 \times 4 = 8$

$a \times a = a^2$

$= 8a^2$

Tip: $a \times a = a^2$, not $2a$.

Task 1

PRACTICE · 9 MIN

BRONZE *Simplify each expression*

1. $4 \times 3p$
2. $2a \times 6b$
3. $5x \times x$
4. $3m \times 2m$
5. $2a \times 3b \times 4c$

★ Simplify $4x \times 2y \times x$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $12p$

2 $12ab$

3 $5x^2$

4 $6m^2$

5 $24abc$

★ $8x^2y$

Examples 3 and 4 — Expanding Brackets

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Expand

$4(2x - 3)$

Example 4

Expand

$x(x + 5)$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$4 \times 2x = 8x$

$4 \times (-3) = -12$

$= \mathbf{8x - 12}$

Tip: multiply every term inside the bracket.

Example 4

$x \times x = x^2$

$x \times 5 = 5x$

$= \mathbf{x^2 + 5x}$

Tip: the term outside can be a letter.

Task 2

PRACTICE · 9 MIN

SILVER

Expand each bracket

1. $3(x + 4)$
2. $5(2y - 1)$
3. $-2(a + 6)$
4. $p(p - 7)$
5. $3x(2x + 5)$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $3x + 12$
2. $10y - 5$
3. $-2a - 12$
4. $p^2 - 7p$
5. $6x^2 + 15x$

Examples 5 and 6 — Expand and Simplify

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5 (2 marks)

Expand and simplify

$3(x + 4) + 2(x - 1)$

Example 6 (2 marks)

Expand and simplify

$5(2y - 3) - 2(y - 4)$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$= 3x + 12 + 2x - 2$

$= 5x + 10$

Tip: expand both brackets, then collect like terms.

Example 6

$= 10y - 15 - 2y + 8$

$= 8y - 7$

Tip: $-2 \times -4 = +8$

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1 Expand and simplify $2(3a + 1) + 4(a + 2)$ (2 marks)

2 Expand and simplify $4(x - 3) - 3(x + 2)$ (2 marks)

3 Expand and simplify $x(x + 3) + 2(x - 5)$ (2 marks)

4 Expand and simplify $3(2p - 1) - 2(p - 6)$ (2 marks)

5 A rectangle has width x cm and length (2x + 3) cm. Write an expression for its area. Expand your answer. (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $6a + 2 + 4a + 8 = 10a + 10$

2. $4x - 12 - 3x - 6 = x - 18$

3. $x^2 + 3x + 2x - 10 = x^2 + 5x - 10$

4. $6p - 3 - 2p + 12 = 4p + 9$

5. $x(2x + 3) = 2x^2 + 3x$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Simplify $3a \times 7b$

SILVER Expand $-4(3x - 2)$

GOLD Expand and simplify $2(4y + 1) - 3(y - 5)$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE 21ab

SILVER $-12x + 8$

GOLD $8y + 2 - 3y + 15 = 5y + 17$

Exam Practice — Multiplying Terms

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

(a) Simplify $4p \times 3q$ (1)

.....

(b) Simplify $2y \times 5y \times y$ (1)

.....

Question 2

(2 marks)

Write an expression for the area of the rectangle. Simplify your answer.

Diagram NOT accurately drawn



.....

Exam Practice — Expanding Brackets

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

(a) Expand $5(3x - 2)$ (1)

.....

(b) Expand $y(y - 7)$ (1)

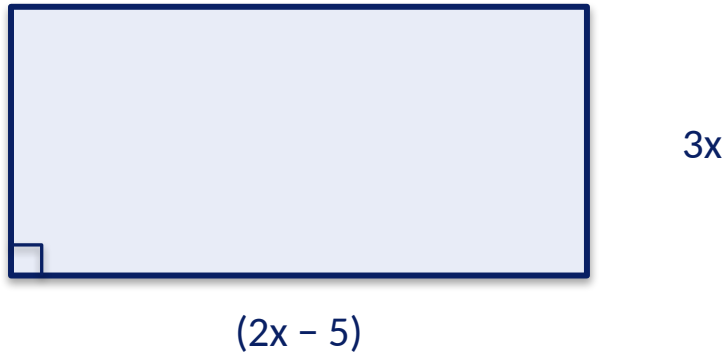
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Question 4

(2 marks)

Show that the area of the rectangle is $6x^2 - 15x$.

Diagram NOT accurately drawn



Exam Practice — Expand and Simplify

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(2 marks)

Expand and simplify

$4(2x + 3) - 3(x - 5)$

.....

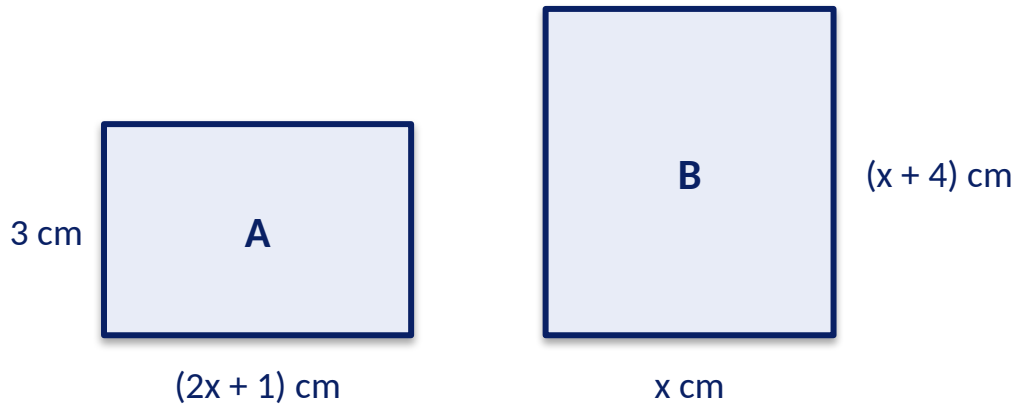
Question 6

(3 marks)

The shape is made from rectangles A and B.

Show that the total area is $x^2 + 10x + 3$.

Diagram NOT accurately drawn



Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 12pq B1

(b) $10y^3$ B1

Tip: multiply numbers, then letters.

Question 2

$3a \times 5b$ M1

$= 15ab$ A1

Tip: area of a rectangle = length × width.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 15x – 10 B1

(b) $y^2 - 7y$ B1

Tip: multiply EVERY term in the bracket.

Question 4

$3x(2x - 5)$ M1

$= 6x^2 - 15x$ (shown) A1

Tip: $3x \times 2x = 6x^2$.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$8x + 12 - 3x + 15$ — M1

$= 5x + 27$ — A1

Tip: $-3 \times -5 = +15$.

Question 6

A: $3(2x + 1) = 6x + 3$ — M1

B: $x(x + 4) = x^2 + 4x$ — M1

$x^2 + 10x + 3$ (shown) — A1

Tip: find each area, then add.